

SEAT No. _____

No of Printed Pages : 2

[50 & A-29]

SARDAR PATEL UNIVERSITY

SYBCA Third Semester EXAMINATION CBCS

17/11/2017 (Friday)

Introduction to Microprocessors- US03EBCA01

Time: 2.00 pm to 4.00 pm

Total Marks: 70

Q.1 Select most appropriate single answer:

[10]

1. An assembler translates the assembly language _____ to the binary codes.
(A) Program (B) mnemonics (C) Code (D) None of these
2. Which of these is not a control flag?
(A) TF (B) IF (C) DF (D) AF
3. AAA stands for _____.
(A) Adjust for Addition (B) Adjust After Addition
(C) ASCII Adjust for Addition (D) None of these
4. DIV stands for _____.
(A) Division (B) Signed Division
(C) Unsigned Division (D) None of these
5. NOT stands for _____.
(A) Negative (B) Nothing
(C) Invert all Byte (D) Invert each Bit
6. ROR stands for Rotate _____ of operand Right.
(A) All Bytes (B) Each Bytes
(C) All Bits (D) Each Bits
7. An END directive ends the _____.
(A) segment (B) program (C) procedure (D) None of these
8. The ASSUME directive is used to tell the assembler the name of the _____ segment it should use for a specific segment.
(A) code (B) Logical (C) data (D) stack
9. The full form of DB directive is _____.
(A) Data Byte (B) Define Bits
(C) Define Byte (D) None of these
10. AND – AND corresponding _____ operands.
(A) Bits of two (B) Bytes of two
(C) Words of two (D) All of these

Q.2 Write Answers in short. Attempt Any Ten.

[20]

1. Define: Decoder and its function in EU.
2. List all conditional flags of 8086 microprocessor.
3. Describe Stack pointer registers.
4. Explain DEC instruction in detail.
5. Explain NOT instruction with example.
6. Explain OR instruction with example.
7. Describe below code.
MOV CL, 02H
ROL AL, CL
8. Explain JMP instruction.
9. Explain SHL instruction.
10. Draw structure of a typical assembly program.
11. Explain END directive.
12. Explain ASSUME directive.

(P.T.O.)

Q.3 Draw the diagram of 8086 internal architecture Also explain Flag Register and General Purpose Registers of 8086 Architecture. [10]

OR

Q.3 [A] Explain Instruction Pointer in detail. [5]

[B] What is an Assembler? Explain the concept of assembler. [5]

Q.4 [A] Explain MOV instruction by taking appropriate examples. [5]

[B] Write a short note on AND instruction. [5]

OR

Q.4 [A] Explain SUB instruction by taking appropriate example. [5]

[B] Write a short note on CMP instruction. [5]

Q.5 [A] Explain SHL instruction with example. [5]

[B] Explain JUMP instruction [5]

OR

Q.5 [A] Explain ROR instruction with example. [5]

[B] Explain LOOPZ instruction with example. [5]

Q.6 [A] Which are the Looping structures? Explain any one in detail. [5]

[B] Explain Assembler directives: ASSUME, DW and LABEL. [5]

OR

Q.6 [A] Write a program for Multiplication of two numbers without using MUL instruction. [5]

[B] Explain IF-THEN-ELSE structure by taking appropriate example. [5]

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